Zero Escape - 9 hours 9 persons 9 doors: Digital Root

Tags:

Time Limit: 1 s Memory Limit: 128 MB
Submission: 699 AC: 429 Score: 85.54

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Description

Junpei was an ordinary college student until he found himself kidnapped and confined to a room abroad some kind of ship. After solving the puzzles keeping him locked in the room, he discovered that 8 other men and women had also been kidnapped including his childhood friend, Akane Kurashiki. Suddenly the voice of someone calling themselves Zero spoke from some speakers overhead:"I mean to have you participate in a game. **The Nonary Game**."

To win the Nonary Game. They had to pass through the Numbered Doors located around the ship to find the 9th door. Three to five participants could scan their bracelets at a door, but their digital root had to match the number of the door to open it. To find the **digital root**, add up the bracelet numbers, than add the numbers form each decimal place of the sum until you have a single-digit number. For example, the digital root of 1 and 2 is 1+2=3, the digital root of 3,5 and 7 is 3+5+7=15=1+5=6.

Input

Your task is to find out the digital roots of the given participants.

The first line is an integer *t*, represents the input has *t*(t≤100) test cases.

For each test case, there are two lines, the first contains an integer *n*(3≤n≤5), represents there are *n* participants. The second line contains *n* numbers *a1,a2...an*(0≤ai≤9, 1≤i≤n), represent the bracelet number of each participant.

Output

For each test case, you have to output a line contains an integer *d* which indicate the digital roots of the participants.

Samples

input Copy

```
2
2
1 2
3
1 2 3
```

output Copy

```
3
```

```
6
```

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